Categroy Theory Reading Group

November 18, 2025

There is a vast variety of motivations to study categories which stem from various domains of formal inquiry, so I choose to first give the definition of a category, and then the motivations will come into light once we start examining particular examples of categories.

0.1 Definition of a Category

In what follows we give the definition (i.e. an axiomatization) of what a *category* is. For readability, we phrase it using some natural language, but for those interested, it is easily expressible formally in either two sorted first order logic or in single sorted first order logic with a predicate for each of the two sorts: *objects* and *arrows*. Without further ado:

Definition 1. A *category* consists of the following data:

- **Objects:** $A, B, C, \dots$
- **Arrows:** $f, g, h, \dots$
- For each arrow f there are given objects $\mathrm{dom}(f), \mathrm{cod}(f)$, called the *domain* and *codomain* of f. We will write $f : A \to B$ or draw:

  $$A \xrightarrow{f} B$$

  to indicate that $A = \mathrm{dom}(f)$ and $B = \mathrm{cod}(f)$.
- Given arrows $f : A \to B$ and $g : B \to C$, i.e. with $\mathrm{cod}(f) = \mathrm{dom}(g)$, there is given an arrow

  $$g \circ f : A \to C$$

  called the *composite* of f and g, we read $g \circ f$ as "f *after* g". The information given here about the domains and codomains can be summarized by the following diagram:

$$
\begin{tikzcd}
 & B \arrow[dr, "g"] & \\
A \arrow[ur, "f"] \arrow[rr, "g \circ f"'] & & C
\end{tikzcd}
$$

- For each object A there is given an arrow

$$1_A : A \to A$$

  called the *identity arrow* of A.

These data are required to satisfy the following laws:

- **Associativity:** $h \circ (g \circ f) = (h \circ g) \circ f$ for all $f : A \to B$, $g : B \to C$, $h : C \to D$.
- **Unit:** $f \circ 1_A = f = 1_B \circ f$ for all $f : A \to B$.

Notice, that the associativity law makes the expression $h \circ g \circ f$ well defined by $h \circ g \circ f := (h \circ g) \circ f = h \circ (g \circ f)$.

Furthermore there is a handy way to express diagrammatically the judgment that some arrows are equal. We say that a diagram *commutes* to express the fact that for any two objets A and B depicted in the diagram, any composition with domain A and codomain B of the arrows depicted is equal to any other such composition. For example to say that the diagram:

$$
\begin{tikzcd}
 & B \arrow[dr, "g"] & \\
A \arrow[ur, "f"] \arrow[rr, "h"'] & & C
\end{tikzcd}
$$

commutes, is to preciseely to say that $g \circ f = h$

We can then, for example, express the information in the Unit axiom by saying that the following diagram commutes:

$$
\begin{tikzcd}
A \arrow[r, "1_A"] \arrow[dr, "f"'] & A \arrow[d, "f"] \arrow[dr, "f"] & \\
 & B \arrow[r, "1_B"'] & B
\end{tikzcd}
$$

Ok, so now we have an abstract definition of what a category is. But what is this abstraction good for? We will partially answer this by considering a few examples of categories.

Besides providing intuition by giving examples, we can also think of this as giving models of category theory. Once again, for those interrested in the formal details, the first example actually gives a second order model since the domain in which objects are interpreted is the *class* of all sets (this is axiomatized by NBG).

0.2 Set

The prototipical but in some sene also least interesting example of a category is the categry of sets and functions **Set**.

- Objects: sets
- Arrows: functions between sets, the domains and codomains are those the domains and codomains in set theoretic sense.
- Composition of Arrows: composition of functions
- Identity Arrows: identity functions

If you are familiar with these notions form set theory, it is easy to see that composition of functions is Associative and the identity functions satisfy the Unit axiom for identity arrows. Thus, **Set** is a category.

Having said so, it is still useful to review the set theoretic definitions of *function* and *composition* to see how awkward they are.

Firstly, a *relation* R between sets A and B is a subset of their cartesian product, i.e. $R \subseteq A \times B = \{(a, b)|\ a \in A \ \ b \in B\}$.

Then a *function* f with domain A and codomain B in set theoretic terms is to be thought of as the set of all input-output pairs, i.e. it is a relation $f \subseteq A \times B$ satisfying the following constraint:

$$\text{For all } a \in A \text{ there exists a unique } b \in B \text{ so that } \ (a, b) \in f$$

This unique b is then denoted as $f(a)$.

The *identity function* of A is defined as:

$$1_A := \{(a, a)|\ a \in A\}$$

Notice however, that the identity relation on A is exactly the same set, i.e. $=_A := \{(a, a)|\ a \in A\}$, thus set theoretically the identity relation and the identity function are the same entity. However conceptually they feel very different. In future session we will see how this is remedied in category theory.

Composition of functions is defined in set theory as follows:

Let A, B, C be sets, and let $f \colon A \to B$ and $g \colon B \to C$ be functions. The *composition* of g with f is the function

$$g \circ f \colon A \to C$$

defined by

$$(g \circ f)(a) := g(f(a)) \quad \text{for all } a \in A.$$

the associativity of composition is then trivial.

However if are to be fully formal, the following akward definition should be stated:

Let $f \subseteq A \times B$ and $g \subseteq B \times C$ be functions. The *composition* $g \circ f$ is the subset of $A \times C$ given by

$$g \circ f = \{(a, c) \in A \times C \mid \exists\, b \in B \text{ such that } (a, b) \in f \text{ and } (b, c) \in g\}.$$

It is not that hard to check then that $g \circ f$ is indeed a function.

0.3 Preorder Categories

We will see how every *preordered set* (*poset*) gives rise to a category.

Definition 2. A *preordered set* is the pair $(P, \leq)$ where P is a set and $\leq \subseteq P \times P$ is a binary relation satisfying for any $a, b, c \in P$:

- (*Reflexivity*) $a \leq a$
- (*Transitivity*) If $a \leq b$ and $b \leq c$, then $a \leq c$

There is then a natural way how $(P, \leq)$ defines a category $\mathbf{P}$ given by:

1. The objects of $\mathbf{P}$ are the elements of P
2. For any $a, b \in P$ there is an arrow $\leq_a^b \colon a \to b$ iff $a \leq b$

Thus between any two objects there are at most one arrow. Transitivity of the partial order makes well defined the compositions $\leq_b^c \circ \leq_a^b = \leq_a^c$, represented by the commutative diagram[1]

$$\begin{tikzcd} & c \arrow[dl, "\leq_b^c"'] \arrow[dd, "\leq_a^c"] \\ b \arrow[dr, "\leq_a^b"'] & \\ & a \end{tikzcd}$$

Reflexivity of the partial order gives the identity arrows $\mathrm{id}_a = \leq_a^a$.

Since given any domain and codomain there is at most one arrow between them the associativity and identity axioms are trivially satisfied.

This example shows us that the we are not forced to think of arrows of a category as morphisms or functions in the usual sense, since the arrows in $\mathbf{P}$ do not denote any evident morphisms.

Another big lesson that thee categories give is that instead of starting out with a set theoretic notion of poset and defining a category we might have proceed with categorical notions altogether. In particular:

Definition 3. A category $\mathcal{C}$ is called a *preorder category* if between any pair of objects there is at most one arrow in any direction.

[1]Note that any Hasse diagram is easily transformed into a commutative diagram by assigning the upward direction to each edge

Thus we might think that it is the preorder categories that give rise to preorders orders and not the other way around.

It is not difficult to recover partial orders this way either. A partially ordered set is a preordered set that additionally satisfies:

- *(Antisymmetry)* If $a \leq b$ and $b \leq a$ then $a = b$

Accordingly then:

Definition 4. A category $\mathcal{C}$ is called a *posetal* if between any pair of objects there is at most one arrow in both directions.

This prevents there being two objects that are distinct but greater or equal than each other.

Let us now consider some specific preorders.

0.3.1 Specifc Preordered Sets

Many preorders are of interrest:

- The transitive and reflexive frames in the semantics of modal logic
- Subsets of a given set with the relation $\subseteq$. This not only a preorder, but a partial order since $A \subseteq B$ and $B \subseteq A$ implies $A = B$.
- Formulas of a logic[2] under (syntactic or semantic) consequence, i.e. $\phi \vdash \psi$, since $\phi \vdash \phi$ and by the cut rule from $\phi \vdash \psi$ and $\phi \vdash \theta$ follows $\phi \vdash \psi$. Note that this is *not* a partial order since we may have distinct equivalent formulas e.g. the formula ϕ is not the formula $\phi \wedge \phi$, but $\phi \vdash \phi \wedge \phi$ and $\phi \wedge \phi \vdash \phi$..
- The divisibility relation between integers
- Etc.

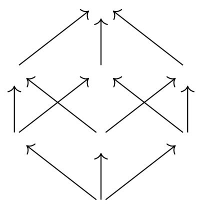
Suitably restricted, some of these preorders produce identically-looking categories[3]. For examaple the the positive divisors of 30 of which there are 8 form an identical-looking preordered set as the subsets of a 3 element set $\{\alpha, \beta, \gamma\}$. Indeed if we pair them up like so

$$\begin{array}{ccl}
\varnothing & \longleftrightarrow & 1 \\
\{\alpha\} & \longleftrightarrow & 2 \\
\{\beta\} & \longleftrightarrow & 3 \\
\{\gamma\} & \longleftrightarrow & 5 \\
\{\alpha, \beta\} & \longleftrightarrow & 6 \\
\{\alpha, \gamma\} & \longleftrightarrow & 10 \\
\{\beta, \gamma\} & \longleftrightarrow & 15 \\
\{\alpha, \beta, \gamma\} & \longleftrightarrow & 30
\end{array}$$

[2]This does not apply to substructural logics, however they were developed by Girard with explicit reference to category theory

[3]In the future this will be defined as an isomorphisms between categories

Then then one set is a subset of another iff the divisors assigned to them devide one another. Thus their category-theoretic diagrams will both look like



This example shows that when we obtain a category from some mathematical structure, category theoretic means may be insufficient to distinguish from which structure did the category come from. So are category theoretic means enough to distinguish, for example the category **Set**? Surprisingly they are, through the category theoretic notion of universal construction. We begin discussing them in the following sections.

1 Terminal Objects

In the category **Set** can we tell apart one set form another jut looking at the arrows? It turns out that we can upto the cadinality of sets. For conider finite sets with sizes m and n, then there will be n^m functions from the former to the latter. So in particular if we know that a set T is such a set that each other has exactly one arrow from it to T, then $|T|^m = 1$ and so the size of T can only be 1. In this way we arrive at a categorical charchterisation of *signleton sets*, i.e. sets conatinig one element.

Definition 5 (Terminal Object)**.** An object T is said to be *terminal* in a category $\mathcal{C}$ iff for any object X of $\mathcal{C}$ there exists a unique arrow from X to T.

Terminal Objects in Set. Terminal objects in **Set** have a special role: they allow us too "look into" what's happening inside of objects. Take any singleton $T = \{*\}$, then there is a natural correspondence between *arrows from* the terminal object and *elements* of a set. Indeed, a function $f : T \to X$ is completely determined by the image of the sole point:

$$f(*) = x \in X.$$

Thus each element $x \in X$ connects uniquely to a function $T \to X$.

Therefore, the terminal object allows us to "look inside" a set X by identifying its elements with maps from T.

This is often expressed by saying that:

"global elements of X are morphisms $T \to X$."

So T acts as a *probe* object: to describe an element of X, we describe a map from the terminal object into X.

Terminal objects in Preorder categories. If we consider what a terminal object t in a preorder category should be, we see that for all $x \in P$ there is an arrow $x \to t$, i.e. $x \leq t$, so t is a *greatest element*. If we require the preorder to be a partial order it is *the* greatest element. Some examples of terminal objects in preorder categories considered above:

- The set S in the preorder category of it's subsets with the ordering $\subseteq$
- Any tautology is a terminal object in the category of formulas ordered by $\vdash$, since tautologies are provable from any premiss.
- The number 0 is divisible by any integer, so it is terminal in the preorder of integers under the divisibility relation

2 Isomorphisms

From the discussion of terminal objects above, we learn that, for example in the category **Set** all terminal objects "look alike" in the sense, that all of them are singletons. Also all of the terminal object in the category of formulas of a logic, also deductively "look alike" in the sense that they are all tautologies. This 'looking alike' however can be given a general definition:

Definition 6 (Isomorphic Objects)**.** In a category $\mathcal{C}$ to objects A and B are said to be isomorphic (denoted $A \cong B$) :iff there exist arrows $f : A \to B$ and $g : B \to A$ such that:

$$f \circ g = 1_B \text{ and } g \circ f = 1_A$$

.

i.e. the diagram commutes:

$$\begin{tikzcd} A \arrow[loop left, "1_A"] \arrow[r, bend right, "f"'] & B \arrow[l, bend right, "g"'] \arrow[loop right, "1_B"] \end{tikzcd}$$

The morphisms f and g are called then *isomorphisms*. One could think of this definitions as guaranteeing that however f scrambles up A mapping it into B, it doesn't loose any structural information about A, since the morphism g can restore all the information back to A. So isomorphic objects can be though of as those accommodating each others structure.

This is made precise, since we will see how the definition given above makes isomorphic objects have the same category theoretic properties. To illustrate, let us go back to terminal objects:

Theorem 1. *Any two terminal objects are isomorphic. Furthermore, if an object is isomorphic to a terminal object it is also terminal.*

Proof. Let T and T'be terminal objects. Since T' is terminal there exists an arrow f from T to T'. Similarly there exists an arrow $g : T' \to T$.

So composing the arrows we get an arrow $g \circ f : T \to T$. But by the axioms of category theory there also exists an arrow $1_T : T \to T$. However by the terminality of T there is a unique arrow from any object X to T, so in particular there is unique arrow from $T \to T$. Thus this unique arrow must be $1_T = g \circ f$. By the same argument, $1_{T'} = f \circ g$. Therefore $T \cong T'$

$$\begin{tikzcd} T \arrow[loop left, "1_T"] \arrow[r, bend right, "f"'] & T' \arrow[l, bend right, "g"'] \arrow[loop right, "1_{T'}"] \end{tikzcd}$$

For the second part, suppose that T is terminal and that $A \cong T$. Then there are isomorphism $f : A \to T$ and $g : T \to A$, this immediatelly implies that there is an arrow from any object X to A, since there must be an arrow $h : X \to T$ to the terminal object T, and so we get $g \circ h : X \to A$.

It remains to prove that there is a unique arrow from any object X to A. Suppose, there are two such arrows $j : X \to A$ and $k : X \to A$ then composing them with f we get $f \circ j : X \to T$ and $f \circ j : X \to T$ which must coincide since there is a unique such arrow, so $f \circ j = f \circ k$. Now let us compose this arrow with g:

$$(g \circ f) \circ j = g \circ (f \circ j) = g \circ (f \circ k) = (g \circ f) \circ k$$

But by definition of isomorphism $g \circ f = 1_A$, and so:

$$1_A \circ j = 1_A \circ k$$

Using the axiom for the identity arrow we finally get:

$$j = k$$

$$\begin{tikzcd} X \arrow[d, bend right, "j"'] \arrow[d, bend left, "k"] \arrow[dr, "f \circ k = f \circ j"] & \\ A \arrow[loop left, "1_A"] \arrow[r, bend left, "f"] & T \arrow[l, bend left, "g"] \arrow[loop right, "1_T"] \end{tikzcd}$$

□

3 Products

In different areas of formal inquiry we repeatedly find a notion that is very much "product-like", for example:

- The cartesian product of sets $A \times B$
- The product of natural numbers $n \cdot m$

- Logical conjunction $\phi \wedge \psi$
- The product-type $A \times B$ of two types A and B in type theory.
- The greatest lower bounds in preordered sets, examples of the include:
  - The intersection of sets $A \cap B$ in the preorder of sets under inclusion $\subseteq$.
  - The minimum of two real numbers $\min(x, y)$ in the usual ordering $\leq$ of $\mathbb{R}$.
  - The greatest common divisor (gcd) of two natural numbers in the preordered set of narutal numbers under the divisibility relation
- The product of two mathematical structures of a certain kind being a mathematical structure of the same kind, examples of these include:
  - Product of two groups $G \times H$
  - Product of two topological spaces $T \times S$
  - Product of two vector spaces $U \times V$

While on the outset it doesn't seem that simple to put a finger on what is it exaclty that unites these notions, it turns out that each of them is *a product* in their respective category.

Definition 7 (Product)**.** An object $A \times B$ of a category $\mathcal{C}$ together with morphisms $\pi_A : A \times B \to A$ and $\pi_B : A \times B \to B$ is a *product* of A and B :iff for any pair of maps $f : X \to A$ and $g : X \to B$ in $\mathcal{C}$ there exists a unique map $\langle f, g\rangle : X \to A \times B$ such that $\pi_A \circ \langle f, g\rangle = f$ and $\pi_B \circ \langle f, g\rangle = g$, i.e. the following diagram[4] commutes:

$$\begin{tikzcd} & X \arrow[dl, "f"'] \arrow[d, dashed, "\langle f,g\rangle"] \arrow[dr, "g"] & \\ A & A \times B \arrow[l, "\pi_A"] \arrow[r, "\pi_B"'] & B \end{tikzcd}$$

It is important to stress here, that there maybe many products of two fixed objects of a category (we will see examples bellow). Therefore, the notation $A \times B$ should be read not as referring to *the* product of A and B, but to *some* product. Thus if we wish to speak of several product at the same time we need to either add subscripts (as will be done in considerations bellow), or distinguish them in notation in some other way.

[4]Note: in this diagram, the dashing of the arrow $\langle f, g\rangle : X \to A \times B$ is meant to indicate that this is the unique arrow such that the diagram commutes. We will make use of this graphical notation in the future too.

Products in Set. Firstly, as the notation suggests, the cartesian product in the category **Set** is indeed a product in the categorical sense. Let us check that.

1. For two sets A and B consider the set their pairs $A \times B = \{(a,b)|\ a \in A\ b \in B\}$

2. Equip it with the natural projections $\pi_A\ A \times B \to A$ and $\pi_B\ A \times B \to B$ defined by:

   - $\pi_A((a,b)) = a$
   - $\pi_B((a,b)) = b$

   for any $(a,b) \in A \times B$

3. To check that it is indeed a product, we need to show that for any two functions $f : X \to A$ and $g : X \to B$ there is a unique map $\langle f, g\rangle : X \to A \times B$ such that $\pi_A \circ \langle f, g\rangle = f$ and $\pi_B \circ \langle f, g\rangle = g$.

   So since the identity of functions is determined by their pointwise values, this means we need to check that for all $x \in X$:

   - $\pi_A(\langle f, g\rangle(x)) = f(x)$
   - $\pi_B(\langle f, g\rangle(x)) = g(x)$

   But this just tells us what the first and second components of $\langle f, g\rangle(x) \in A \times B$ are so for each x we obtain a definition of $\langle f, g\rangle(x)$, namely:

   $$\langle f, g\rangle(x) := (f(x), g(x))$$

4. Since functions are determined poitnwise, we have also shown that the function $\langle f, g\rangle$ satisfying the condition need for the product is unique.

Philosophical point. So we have shown that the cartesian product in **Set** is a product in the category theoretic sense. Importantly, however, we have glossed over what exactly *a pair* (a,b) is. Since pairs are not part of the primitive vocabulary of set theory, if one whishes to have a set theoretic treatment of them, one has to come up with a definition, and indeed people have done so, for example:

- Norbert Wiener. $(a,b)_W := \{\{\{a\}, \emptyset\}, \{\{b\}\}\}$
- Kazimierz Kuratowski. $(a,b)_K := \{\{a\}, \{a,b\}\}$

Evidently, these definitions conflict e.g. in Kuratowski's definition a is an element both elements of (a,b), while in Wieners it is not. Accordingly, depending on what definition we use, we get an (extensionally) different definition of cartesian product:

- $A \times_W B := \{(a,b)_W|\ a \in A\ b \in B\} = \{\ \{\{\{a\}, \emptyset\}, \{\{b\}\}\}|\ a \in A\ b \in B\}$

- $A \times_K B := \{(a,b)_K |\ a \in A\ b \in B\} = \{\ \{\{a\},\{a,b\}\} |\ a \in A\ b \in B\}$

But what are we to make of these conflicting definitions? One line of approach, would be to follow Benacerraf in his influential Benacerraf 1965 and argue that these conflicts tell us that neither of the definiens is what pairs are.

In a somewhat contrary approach, Quine in Quine 1960 takes this exact example of pairs as “philosophical paradigm” of “analysis” or “explication”. Quine's point seems to be here that before a definition like the two above is given “ordered pair” is a “defective noun”, once the definition given it is given an explication that fixes this “defectiveness” introducing however some *don't-cares*. These don't-cares are exactly the conflicting properties described above, they are, so to speak, the price to pay if we wish to explicate the notion. But what are the “cares” then? Well, for Quine, the main “care” of pairs and the property that both of these definitions share is:

$$(a,b) = (x,y) \text{ iff } a = x \ \&\ b = y$$

Accepting Quine's account, the main issue when giving such an explication is then to sort out what the cares and the don't-cares are. It is here, in my opinion, that the advantage of category theoretic definitions is clearly seen, for they do this sorting out automatically!

To illustrate, let us return to the two cartesian products induced by the two definitions of pair:

- $A \times_W B$
- $A \times_K B$

while these are different sets, they are both products in the category theoretic sense if we equip them with the natural projections:

- $\pi_A^W(\{\{\{a\},\emptyset\},\{\{b\}\}\}) := a$ and $\pi_A^W(\{\{\{a\},\emptyset\},\{\{b\}\}\}) := b$
- $\pi_A^K(\{\{a\},\{a,b\}\}) := a$ and $\pi_B^K(\{\{a\},\{a,b\}\}) := b$

Thus we might rephrase, that the “care” that makes both of the sets above cartesian products, is exactly the fact that they are products in the category theoretical sense. We can then define pairs to be global elements of products, giving us the “care” of pairs that Quine identifies.

Products in Preorder Categories. Consider a preorder category **P** obtained from a preorder $(P, \leq)$, what is a product of $a, b \in P$ in this category?

Well. firstly, a product $a \times b$ has arrows into both a and b, so in other words $a \times b \leq a$ and $a \times b \leq b$, so it is a *lower bound* of a and b.

On the other hand, for any x with arrows $x \to a$ and $x \to b$, that is for an x that is a lower bound of a and b, there is an arrow from x to $a \times b$. This means that $x \leq a \times b$, for any other lower bound x of a and b. Thus, $a \times b$ is a *greatest* *lower bound* of a and b. If the preorder is a partial order it becomes unique and is usually denoted by $a \wedge b$.

Some examples of greatest lower bounds:

- In the preorder category of subsets of a set, the intersection $A \cap B$ is the greatest lower bound of A and B
- In the preorder category of natural numbers under the divisibility relation it is the greatest common denominator.
- In the category of formulas of a logic, the conjunction of two formulas is a product. Indeed by the conjunction elimination rules $\phi \wedge \psi \vdash \phi$ and $\phi \wedge \psi \vdash \psi$, but also if $\theta \vdash \phi$ and $\theta \vdash \phi$ using the conjunction introduction rule we get $\theta \vdash \phi \wedge \psi$. In this way we see that the usual introduction/elimination rules in proof theory mimic the definition of category theoretic product.

4 Functors

In section 0.3.1 we have discussed two categories that we have informally said to "look alike". The aim of this section is to setup formal tools that allow us to study how different categories relate to each other. All of this will be done trough the notion of *functor*, which is to be though of as homomorphism between categories. Thus, formally:

Definition 8. A *functor* $F : \mathcal{C} \to \mathcal{D}$ from a category $\mathcal{C}$ to a category $\mathcal{D}$ is a map:

- Assigning to each object A of $\mathcal{C}$ an object $F(A)$ of D
- Assigning to each arrow f of $\mathcal{C}$ an arrow $F(f)$ of $\mathcal{D}$

Such that the category theoretic structure is preserved, i.e.:

- $F(\text{dom}(f)) = \text{dom}(F(f))$ and $F(\text{cod}(f)) = \text{cod}(F(f))$
- $F(1_A) = 1_{F(A)}$
- $F(f \circ g) = F(f) \circ F(g)$

Diagramatically depicted as follows:

$$\begin{tikzcd} & \mathcal{C} \arrow[rrrr, bend left, "F"] & & & & \mathcal{D} & \\ & B \arrow[dr, "g"] & & & & F(B) \arrow[dr, "F(g)"] & \\ A \arrow[ur, "f"] \arrow[rr, "g\circ f"'] & & C & & F(A) \arrow[ur, "F(f)"] \arrow[rr, "F(g\circ f)=F(g)\circ F(f)"'] & & F(C) \end{tikzcd}$$

Before developing this notion further let us consider a few examples

Example 4.1. Every category $\mathcal{C}$ has an identity functor $1_{\mathcal{C}}$ defined by:

- $1_{\mathcal{C}}(A) = A$
- $1_{\mathcal{C}}(f) = f$

It is trivial that this is a well defined functor

Example 4.2. Let $\mathbf{P}$ and $\mathbf{Q}$ be two preorder categories constructed from preordered sets $(P, \leq_P)$ and $(Q, \leq_Q)$ respectively, what is a functor $F : \mathbf{P} \to \mathbf{Q}$ between them?

Well any arrow $a \to b$ in $\mathbf{P}$ just means that $a \leq_P b$, thus when this arrow gets mapped to $F(a) \to F(b)$, this means that $F(a) \leq_Q F(b)$. In other words:

$$a \leq_P b \Rightarrow F(a) \leq_Q F(b)$$

But this is just the definition of a monotonic function! Thus functors between preorder categories can be though of as monotonic functions.

Example 4.3 (Interpretation of theories)**.** Halvorson in Halvorson 2012 gives a quick definition of what is an intepretation of one theory into another. Viewed appropriately, these interpretation are functors.

Let T and T' be two theories, let $Form(T)$ and $Form(T')$ be the sets of all formulas in the languages of T and T' respectivelly. Then each theory induces a syntactic entailment relation between its formulas, i.e. $\vdash_T$ and $\vdash_{T'}$ respectivelly. Thus we have two partial orders that give rise to two preorder categories:

- The formulas $Form(T)$ under under $\vdash_T$; gives a category $\mathbf{T}$
- The formulas of $Form(T')$ under under $\vdash_{T'}$; gives a category $\mathbf{T'}$

Then, by Halvorson, an *interpretation* F is a map from $Form(T)$ to $Form(T')$ such that:

$$\text{If } \psi \vdash_T \phi, \text{ then } F(\psi) \vdash_{T'} F(\phi)$$

Evidently, these interpretaions are functors between $\mathbf{T}$ and $\mathbf{T}'$ just as the example considered above.

Furthermore Halvorson uses these intepretations to define *definitional equivalence*, we will see that in the language of functors this amounts to equivalence of categories.

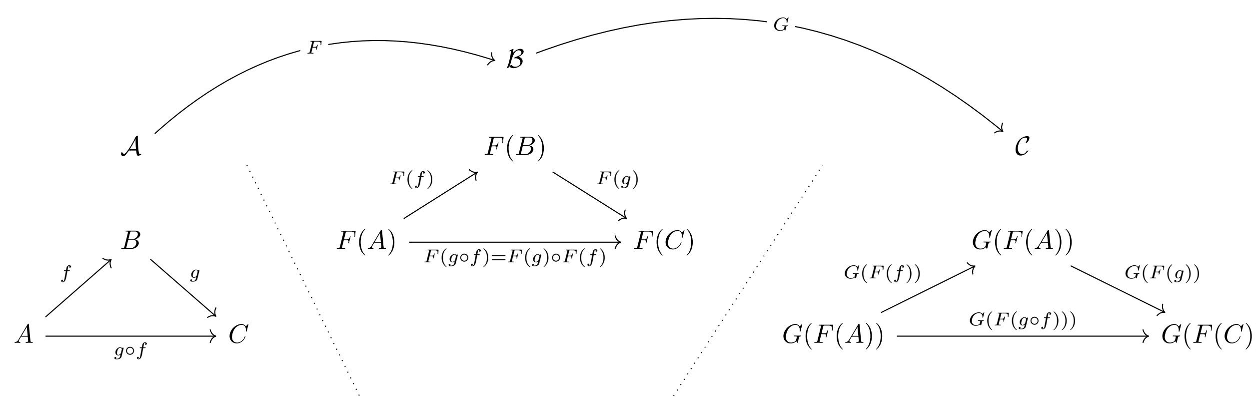
4.1 A Category of Categories?

The notation $F : \mathcal{C} \to \mathcal{D}$ and $1_{\mathcal{C}}$ suggest that functors themselves are to be thought of as arrows. This is indeed so, and the last piece that we need is to define how functors are to compose, which is done in a natural way:

Definition 9. Given two functors $F : \mathcal{A} \to \mathcal{B}$ and $G : \mathcal{B} \to \mathcal{C}$ their composition $G \circ F : \mathcal{A} \to \mathcal{C}$ is defined for obejcts A and arrows f of $\mathcal{A}$ by:

- $G \circ F(A) = G(F(A))$
- $G \circ F(f) = G(F(f))$

It is easy to check that $G \circ F$ is indeed also a functor. Furthermore, compostion of functors is associative, i.e. $(F \circ G) \circ H = F \circ (G \circ H)$



Thus each category has an identity functor, functors have domains and codomains, they compose associatively. So is there then a category in the background, presumably a category whose objects themselves are categories, and arrows are functors?

The answer is quite subtle. Consider for example:

Definition 10. The category **PreOrd** is a category whose:

- Objects are preorder categories
- Arrows are functors between preorder categories (i.e. monotonic functions)

This indeed a category whose objects are categories and arrows are functors. The looming question is however, is there a *category of all categories*, thus a category which has itself as an object? The answer to this will depend on our views on the philosophy and foundations of mathematics, thus let us postpone this for the moment.

4.2 Isomorphism of categories

In section 2 we have introduced what it means for two objects of a category to be isomorphic, we can now extend this notion to two categories being isomorphic, independently of whether we think of them as living in some larger category.

Definition 11. Two categories $\mathcal{C}$ and $\mathcal{D}$ are said to be isomorphic (denoted by $\mathcal{C} \cong \mathcal{D}$), if there exist functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ such that $G \circ F = 1_{\mathcal{D}}$ and $F \circ G = 1_{\mathcal{C}}$

Thus we have two ways how categories can relate:

1. Equality. $\mathcal{C} = \mathcal{D}$
2. Isomorphism. $\mathcal{C} \cong \mathcal{D}$

In section 0.3.1 we had an example of two categories that are not equal but are isomorphic. Furthermore, trivially, equal categories are isomorphic (through the identity functor). Thus isomorphism, is a weaker relation than equality. It turns out however that even isomorphism is also sometimes too strung and an even weaker relation is of interest in those cases, the relation being *equivalence* of categories $\mathcal{C} \simeq \mathcal{D}$. As we work toward this notion let us consider a few more examples of functors.

Example 4.4 (The cardinality functor)**.** Two sets (so objects of the category **Set**) A and B are said to be *equinumorous*, or to have the same *cardinality* (denoted $A \cong B$), if there exists a bijection between them, which is precisely an isomorphism in **Set**.

But suppose we don't want to speak speak of two sets, but of *the cardinality* of a set A. In set theory it is defined to be the least ordinal that is equinumerous with A, i.e.:

$$Card(A) = |A| \ := \ \min\{\alpha \in \mathrm{Ord} : A \cong \alpha\}.$$

Where the (finite) ordinals are constructed as follows:

$$
\begin{aligned}
0 &= \varnothing, \\
1 &= \{0\}, \\
2 &= \{0,1\}, \\
3 &= \{0,1,2\}, \\
4 &= \{0,1,2,3\}, \\
&\vdots \\
n &= \{0,1,\ldots,n-1\}, \\
&\vdots
\end{aligned}
$$

This definition then extends to transfinite ordinals, but the finite ones will suffice to illustrate our point, since we will view the cardinality as a functor $Card : \mathbf{Set} \to \mathbf{Set}$. So far, $Card$ is definied on the objects of **Set**, e.g. $Card(\{\square, \star\}) = 2 = \{0,1\}$, but how is to be defined for the arrows. i.e. the functions?

Well consider, any function $f : A \to B$. By definition of cardinality:

- A is isomorphic (equinumerous) to $Card(A)$, so there are ismorphisms $\alpha_A : A \to Card(A)$ and $\alpha_A^{-1} : Card(A) \to A$, such that $\alpha_A \circ \alpha_A^{-1} = 1_A$ and $\alpha_A^{-1} \circ \alpha_A = 1_{Card(A)}$
- B is isomorphic (equinumerous) to $Card(B)$, so there are ismorphisms $\alpha_B : B \to Card(B)$ and $\alpha_B^{-1} : Card(B) \to B$, such that $\alpha_B \circ \alpha_B^{-1} = 1_B$ and $\alpha_B^{-1} \circ \alpha_B = 1_{Card(B)}$

$$
\begin{tikzcd}
A \arrow[d, bend right, "\alpha_A"'] \\
Card(A) \arrow[u, bend right, "\alpha_A^{-1}"']
\end{tikzcd}
\qquad
\begin{tikzcd}
B \arrow[d, bend right, "\alpha_B"'] \\
Card(B) \arrow[u, bend right, "\alpha_B^{-1}"']
\end{tikzcd}
$$

We can use these isomorphisms then to define $Card(f) : Card(A) \to Card(B)$ as

$$Card(f) := \alpha_B \circ f \circ \alpha_A^{-1}$$

i.e. by going around the diagram

$$
\begin{tikzcd}
A \arrow[rr, "f"] & & B \arrow[d, bend right, "\alpha_B"'] \\
Card(A) \arrow[u, bend right, "\alpha_A^{-1}"'] \arrow[rr, dashed, "Card(f)"'] & & Card(B)
\end{tikzcd}
$$

It is not difficult to check that this indeed defines a functor since, for a composition of f with a function $g : B \to C$ we get:

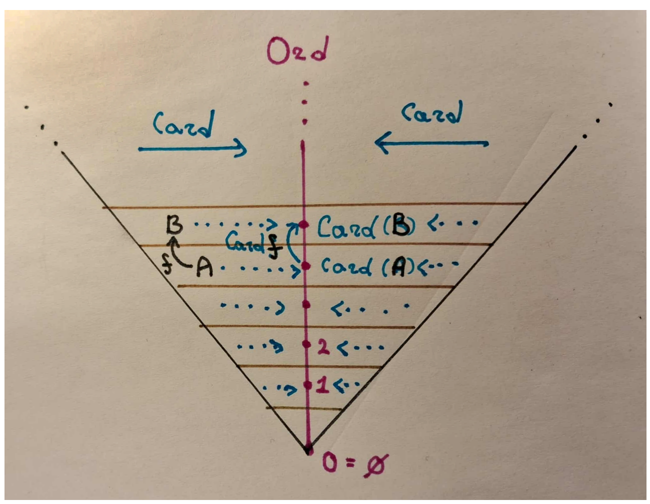
$$
\begin{aligned}
\mathrm{Card}(g \circ f) &= \alpha_C \circ g \circ f \circ \alpha_A^{-1} \\
&= \alpha_C \circ g \circ 1_B \circ f \circ \alpha_A^{-1} \\
&= \alpha_C \circ g \circ \alpha_B^{-1} \circ \alpha_B \circ f \circ \alpha_A^{-1} \\
&= (\alpha_C \circ g \circ \alpha_B^{-1}) \circ (\alpha_B \circ f \circ \alpha_A^{-1}) \\
&= \mathrm{Card}(g) \circ \mathrm{Card}(f).
\end{aligned}
$$

diagrammatically:

$$
\begin{tikzcd}
A \arrow[r, "f"] & B \arrow[r, "g"] \arrow[d, bend right, "\alpha_B"'] & C \arrow[d, bend right, "\alpha_C"'] \\
Card(A) \arrow[u, bend right, "\alpha_A^{-1}"'] \arrow[r, "Card(f)"'] \arrow[rr, bend right, "Card(g \circ f))"'] & Card(B) \arrow[u, bend right, "\alpha_B^{-1}"'] \arrow[r, "Card(g)"'] & Card(C)
\end{tikzcd}
$$

Thus we have defined how $Card$ acts on both objects and arrows, we have checked that it acts on composition of arrows as a functor should, checking the identity is trivial, so $Card : \mathbf{Set} \to \mathbf{Set}$ is indeed a functor.

In fact since for each set the functor return an ordinal, we may view it as $Card : \mathbf{Set} \to \mathbf{Ord}$, where **Ord** is the category whose objects are ordinals and whose arrows are all the functions between ordinals. This induces a picture of the category **Set** that you might be familiar with from set theory:



Sorry for the hand drawn picutre, but it takes forever to typeset these things in latex

So the functor $Card$ may be viewed as sqeueezing in the category set onto the ordinals. This means that sets of the same cardinality get glued together, and so there will be no inverse to . On the other hand $Card$ doesn't loose that much information, since the most important thing about a set is it's cardinality and this is preserved. In the following section we will fromaly capture this by saying that **Set** and **Ord** are *equivalent* categories.

5 Natural Transformations

To recap we have thus far considered two levels of categories:

- Categories *per se*, i.e. where there is no further structure to the objects and arrows
- Categories in which objects themselves are categories and arrows are functors

The aim of this section is to go one step further and to consider categories whose objects are functors and whose maps are *natural transformations*. This will allow us to define what it means for categories to be equivalent, give a general definition of *universal construction* of which we have considered terminal objects and products, as well as further usefull applications.

Definition 12 (Natural transformation)**.** Given two functors $F : \mathcal{A} \to \mathcal{B}$ and $G : \mathcal{A} \to \mathcal{B}$ a *natural transformation* $\alpha : F \to G$ is a collection of arrows $\{\alpha_A\}_{A \in Obj(\mathcal{A})}$ from the category B such that, for any arrow $f : A \to B$ in $\mathcal{A}$

$$\alpha_B \circ F(f) = G(f) \circ \alpha_A$$

i.e. diagramatically, it must make the diagram on the left commute:

$\mathcal{A}$ $\qquad\qquad$ $\mathcal{B}$

$$\begin{tikzcd} A \arrow[r, "f"] & B \end{tikzcd} \qquad\qquad \begin{tikzcd} F(A) \arrow[r, "F(f)"] \arrow[d, "\alpha_A"'] & F(B) \arrow[d, "\alpha_B"] \\ G(A) \arrow[r, "G(f)"] & G(B) \end{tikzcd}$$

F $\qquad$ G

Just to reiterate, α is an assignement for each object A of $\mathcal{A}$ of an arrow form $\mathcal{B}$ such that the above requirement is satsified. These arrows α_A are called *components* of the natural transformation α.

We have actually already discussed an example of a non-trivial natural transformation while talking about the cardinality functor $Card : \mathbf{Set} \to \mathbf{Set}$. There the isomorphisms α_A between a set A and its cardinality $Card(A)$ are exactly components of a natural transformation α between the identity functor $1_{\mathbf{Set}}$ and the functor $Card$.

5.1 Category of Functors

Unsurprisingly each functor F has an identity natural transformation $1_F : F \to F$ given by $(1_F)_A := 1_{F(A)}$, this of course satsfies the definition:

$$\begin{tikzcd}
F(A) \arrow[r, "F(f)"] \arrow[d, "1_{F(A)}"'] & F(B) \arrow[d, "1_{F(B)}"] \\
F(A) \arrow[r, "F(f)"] & F(B)
\end{tikzcd}$$

Furthermore, one can define naturally how natural transformations compose.

Definition 13 (Composition of natural transformations)**.** Given two natural transformations $\alpha : F \to G$ and $\beta : G \to H$ where G, F and H are functors from $\mathcal{A}$ to $\mathcal{B}$, their composition $\beta \circ \alpha : F \to H$ is defined by:

$$(\beta \circ \alpha)_A := \beta_A \circ \alpha_A$$

To check that $\beta \circ \alpha$ is indeed a natural transformation we need to chechk that the big rectangle in the following diagram commutes:

$$\begin{tikzcd}
F(A) \arrow[r, "F(f)"] \arrow[d, "\alpha_A"'] & F(B) \arrow[d, "\alpha_B"] \\
G(A) \arrow[r, "G(f)"] \arrow[d, "\beta A"'] & G(B) \arrow[d, "\beta_B"] \\
H(A) \arrow[r, "H(f)"] & H(B)
\end{tikzcd}$$

But this is trivial, since we know that the two small squares commute in virtue of α and β being natural transformations themselves. Adding another such square for a natural transformation $\gamma : H \to I$ we see that the composition of natural transformations is assocaitive.

To summarise:

- All functors have an identity natural transformation

- Composition of two natural trasformations between functors $F, G, H : \mathcal{A} \to \mathcal{B}$ is also a natural transformation. Furthermore, this composition is associative

This is all we need for a category, thus:

Definition 14 (Functor category)**.** For two categories $\mathcal{A}$ and $\mathcal{B}$ their *functor category* (denoted $[\mathcal{A}, \mathcal{B}]$) is a category whose:

- Objects are functors $F : \mathcal{A} \to \mathcal{B}$
- Arrows are natural transformations $\alpha : F \to G$ between these functors

Since this is category, we can import any machinery developed for categories, for example two functors F and G are said to be *ismorphic* (denoted $F \cong G$) when they are ismorphic as objects of the category $[\mathcal{A}, \mathcal{B}]$, i.e. when there exist natural transformations $\alpha : F \to G$ and $\beta : G \to F$ such that $\beta \circ \alpha = 1_F$ and $\alpha \circ \beta = 1_G$

We are finally set to define what it means for categories to be *equivalent*.

Definition 15 (Equivalence of categories)**.** An equivalence of categories is a pair of functors $F : \mathcal{A} \to \mathcal{B}$ and $G : \mathcal{B} \to \mathcal{A}$ such that $G \circ F \cong 1_A$ and $F \circ G \cong 1_B$.

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